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$$T_4(f, 0)(x) \text{ voor } f(x) = \frac{1}{1+x}$$

$$T_n(f, a)(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k$$

$$T_4(f, 0)(x) = \sum_{k=0}^4 \frac{\left(\frac{1}{1+x}\right)^{(k)}(0)}{k!} (x-0)^k$$

$$\left(\frac{1}{1+x}\right)' = (1+x)^{-2}$$

$$\left(\frac{1}{1+x}\right)'' = -2(1+x)^{-3}$$

$$\left(\frac{1}{1+x}\right)''' = 6(1+x)^{-4}$$

$$\left(\frac{1}{1+x}\right)^{iv} = -24(1+x)^{-5}$$

$$\begin{aligned} \Rightarrow T_4(f, 0)(x) &= \frac{\left(\frac{1}{1+x}\right)^{(0)}(0)}{0!} x^0 + \frac{\left((1+x)^{-2}\right)^{(0)}(0)}{1!} x^1 + \frac{\left(-2(1+x)^{-3}\right)^{(0)}(0)}{2!} x^2 + \\ &\frac{\left(6(1+x)^{-4}\right)^{(0)}(0)}{3!} x^3 + \frac{\left(-24(1+x)^{-5}\right)^{(0)}(0)}{4!} x^4 \\ &= \frac{1}{1} x^0 + \frac{1^{-2}}{1} x^1 + \frac{-2}{2} x^2 + \frac{6}{6} x^3 + \frac{-24}{24} x^4 \\ &= 1 + x - x^2 + x^3 - x^4 \end{aligned}$$

References